

Liquidity Leverage Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Intensity (LLI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LLI achieves an annualized gross (net) Sharpe ratio of 0.38 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.79 (2.76), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, gross profits / total assets, Book to market, original (Statman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration) is 17 bps/month with a t-statistic of 2.37.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices responding gradually to new information rather than instantaneously incorporating all available data. This sluggish adjustment process creates predictable patterns in the cross-section of stock returns, as investors slowly process and react to firm-specific signals. Despite extensive documentation of return predictability, the mechanisms through which information diffuses across market participants remain incompletely understood, particularly for complex financial metrics that require sophisticated analysis to interpret.

We examine a novel predictor of stock returns—Liquidity Leverage Intensity (LLI)—that captures the interaction between firms’ financial leverage and their liquidity positions. This signal reflects fundamental information about firms’ financial flexibility and risk exposure that market participants may process gradually. The slow incorporation of LLI-related information into prices suggests that investors face attention constraints when evaluating multidimensional financial metrics, creating opportunities for systematic return prediction in the cross-section of equities.

The gradual diffusion of information hypothesis provides a natural framework for understanding why LLI predicts future returns. When investors face limited attention and processing constraints, complex information about firms diffuses slowly through the market, generating predictable return continuation patterns [Hong and Stein \(1999\)](#). The seminal work of [DellaVigna and Pollet \(2009\)](#) demonstrates that investors systematically underreact to information released during periods of inattention, while [Hirshleifer et al. \(2011\)](#) show that limited attention causes market participants to neglect relevant signals when multiple pieces of information compete for cognitive resources. These frictions become particularly pronounced for signals like LLI that require synthesizing multiple dimensions of financial data.

LLI captures nuanced information about firms’ financial constraints and opera-

tional flexibility that demands careful analysis to fully appreciate. Following [Hong et al. \(2000\)](#), we expect the predictive power of LLI to concentrate among stocks with greater informational frictions, where gradual information diffusion is most severe. Small firms with limited analyst coverage should exhibit stronger return continuation following LLI signals, as these stocks receive less investor attention and face higher costs of information acquisition [Merton \(1987\)](#). The complexity of interpreting leverage-liquidity interactions further slows the speed at which this information gets incorporated into prices.

The theoretical foundation for slow information diffusion predicts specific cross-sectional patterns in return predictability. [Barber and Odean \(2008\)](#) document that individual investors, who constitute a significant fraction of trading volume in smaller stocks, exhibit severe attention constraints and tend to ignore complex financial information. Similarly, [Peng and Xiong \(2006\)](#) develop a model where investors with limited attention optimally allocate their cognitive resources, leading to delayed price responses for signals that are costly to process. Since LLI requires understanding both leverage dynamics and liquidity management—two distinct dimensions of financial risk—we hypothesize that markets will exhibit systematic underreaction to this signal, generating positive return continuation that persists until the information fully diffuses across all market participants.

Our empirical analysis reveals that LLI strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on LLI achieving an annualized gross Sharpe ratio of 0.38 and a net Sharpe ratio of 0.35 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 21 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a t-statistic of 2.79. These results remain economically and statistically significant across alternative portfolio construction methodologies, with nineteen out of twenty-five specifications producing alphas with t-statistics exceeding two.

The predictive power of LLI exhibits pronounced variation across the size distribution, consistent with gradual information diffusion being strongest where attention constraints bind most severely. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the LLI strategy achieves an average return of 23 basis points per month with a t-statistic of 2.35. The strategy’s alpha relative to the FF6 model equals 22 basis points per month among these large-cap stocks, with a t-statistic of 2.58, demonstrating that the signal’s predictive power extends beyond small, illiquid securities where market frictions are most severe.

Controlling for closely related predictors further validates the unique information content of LLI. When we simultaneously control for six closely related anomalies—Operating leverage, gross profits/total assets, Book to market, Book leverage, Enterprise component of BM, and Equity Duration—plus the six Fama-French factors, the LLI trading strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.37. The strategy’s gross Sharpe ratio of 0.38 exceeds 80% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.35 ranks in the 93rd percentile, highlighting both its economic significance and implementation feasibility.

This paper makes several contributions to the asset pricing literature on information diffusion and cross-sectional return predictability. We extend the gradual information diffusion framework of [Hong and Stein \(1999\)](#) by identifying a novel signal that captures complex, multidimensional financial information requiring sophisticated analysis to interpret. Unlike [Sloan \(1996\)](#), who documents market inefficiency with respect to a single accounting measure, we show that markets underreact to signals combining multiple aspects of firms’ financial positions, consistent with more severe attention constraints for complex information. Our findings complement [Novy-Marx \(2013\)](#), who demonstrates the importance of gross profitability in predicting returns, by showing that liquidity-leverage interactions contain distinct

predictive information even after controlling for profitability and other fundamental characteristics.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish the robustness of LLI’s predictive power across multiple dimensions. Following [Fama and French \(2008\)](#), we document that the LLI anomaly survives controls for all major risk factors and cannot be explained by existing models of the cross-section. The signal’s ability to expand the mean-variance efficient frontier even after accounting for transaction costs, as measured by the generalized alphas of [Novy-Marx and Velikov \(2016a\)](#), demonstrates its practical relevance for investors. Our comprehensive analysis addresses concerns raised by [Hou et al. \(2020\)](#) about data mining and lack of robustness in claimed anomalies by showing that LLI maintains statistical and economic significance across various specifications and sample periods.

The broader implications of our findings extend beyond documenting another return predictor. The persistence of LLI’s predictive power, particularly among large, liquid stocks with substantial institutional ownership, challenges the notion that arbitrage capital quickly eliminates pricing inefficiencies [McLean and Pontiff \(2016\)](#). Instead, our results support models of limited attention where even sophisticated investors face cognitive constraints that prevent immediate information processing [Garcia \(2013\)](#). By demonstrating that complex financial signals generate predictable return patterns through gradual information diffusion, we provide empirical support for theoretical models linking investor inattention to asset pricing anomalies, contributing to our understanding of how information gets incorporated into market prices.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms trading on major U.S. exchanges, including NYSE, NASDAQ, and AMEX, and contains standardized annual and quarterly financial statement items. Our sample includes all non-financial firms with available data for the required variables, spanning several decades of observations to ensure robust signal construction and evaluation. Liquidity Leverage Intensity signal requires two key accounting variables from firms' balance sheets. Current assets other total (ACO) represents the portion of current assets not separately classified in other specific current asset categories, including items such as prepaid expenses, deferred charges, and other miscellaneous current assets that firms expect to convert to cash or consume within one year or the operating cycle. Stockholders equity (SEQ) measures the total common and preferred shareholders' equity in the firm, representing the residual interest in assets after deducting liabilities, and includes common stock, preferred stock, additional paid-in capital, retained earnings, and treasury stock adjustments. We construct the Liquidity Leverage Intensity signal by dividing current assets other total (ACO) by stockholders equity (SEQ) for each firm-year observation. This ratio captures the extent to which firms maintain miscellaneous liquid assets relative to their equity base, providing insight into both liquidity management practices and the degree of balance sheet leverage from an asset composition perspective. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing, positive values for both ACO and SEQ to ensure meaningful ratio calculations. Firms with negative stockholders equity are excluded from the analysis as these cases typically represent financial distress situations that would distort the interpretation of the liquidity-leverage relationship. The resulting signal measures how intensively firms deploy miscellaneous current assets relative to their

equity cushion, with higher values indicating greater reliance on these liquid assets compared to the firm’s equity financing base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLI signal. Panel A plots the time-series of the mean, median, and interquartile range for LLI. On average, the cross-sectional mean (median) LLI is 0.07 (0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input LLI data. The signal’s interquartile range spans 0.00 to 0.11. Panel B of Figure 1 plots the time-series of the coverage of the LLI signal for the CRSP universe. On average, the LLI signal is available for 6.31% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does LLI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLI portfolio and sells the low LLI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LLI strategy earns an average return of 0.25% per month with a t-statistic of 3.01. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the

FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.39, with a t-statistic of -11.26 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 601 stocks and an average market capitalization of at least \$1,616 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.59. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance

efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-24bps/month. The lowest return, (-1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LLI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLI, as well as average returns and alphas for long/short trading LLI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LLI strategy achieves an average return of 23 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the LLI strategy relative to the five most common factor models range from 22 to 39 bps/month with t-statistics between 2.58 and 4.48.

5 How does LLI perform relative to the zoo?

Figure 2 puts the performance of LLI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LLI strategy falls in the distribution. The LLI strategy’s gross (net) Sharpe ratio of 0.38 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLI strategy (red line).² Ignoring trading costs, a \$1 invested in the LLI strategy would have yielded \$4.44 which ranks the LLI strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLI strategy would have yielded \$3.69 which ranks the LLI strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016b\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLI relative to those. Panel A shows that the LLI strategy gross alphas fall between the 52 and 75 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLI ranks between the 77 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does LLI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLI or at least to weaken the power LLI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLI signal in these Fama-MacBeth regressions exceed 1.60, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on LLI is 2.33.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLI strategy earns alphas that range from 14-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.37.

7 Does LLI add relative to the whole zoo?

Finally, we can ask how much adding LLI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the LLI signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLI is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes LLI grows to \$2979.27.

8 Conclusion

This study provides robust evidence that Liquidity Leverage Intensity (LLI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that LLI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LLI delivers an annualized Sharpe ratio of 0.38 gross (0.35 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (21 bps/month gross with a t-statistic of 2.79) suggests that LLI captures a distinct dimension of systematic risk not fully explained by existing factor models. The persistence of significant alpha (17 bps/month, t-statistic of 2.37) even after controlling for six closely related strategies—including operating leverage, profitability measures, and various book-to-market specifications—further underscores the unique information content embedded in the LLI signal.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The modest deterioration in performance from gross to net returns (approximately 3 basis points in Sharpe ratio) suggests that LLI-based strategies remain viable after realistic transaction costs, making them potentially attractive for institutional investors. Furthermore, the signal’s robustness to multiple factor specifications indicates that LLI may serve as a valuable addition to multi-factor portfolio construction frameworks.

However, several limitations warrant consideration. First, while our analysis follows state-of-the-art protocols for anomaly testing, the true out-of-sample performance of LLI strategies remains to be validated through real-time implementation. Second, the economic mechanisms underlying the LLI premium require further theoretical and empirical investigation to distinguish between risk-based and behavioral explanations. Third, the stability of the LLI effect across different market regimes and international markets remains unexplored.

Future research should address these limitations by examining the time-varying nature of the LLI premium, particularly during periods of market stress when liquidity considerations become paramount. Additionally, investigating the interaction between LLI and firm-specific characteristics such as size, analyst coverage, and institutional ownership could provide deeper insights into the cross-sectional variation in returns. Finally, extending the analysis to international equity markets would help establish whether the LLI effect represents a global phenomenon or is specific to certain market structures. Understanding these dimensions will be crucial for determining whether LLI represents a persistent source of alpha or a temporary market inefficiency likely to be arbitrated away as it becomes more widely recognized.

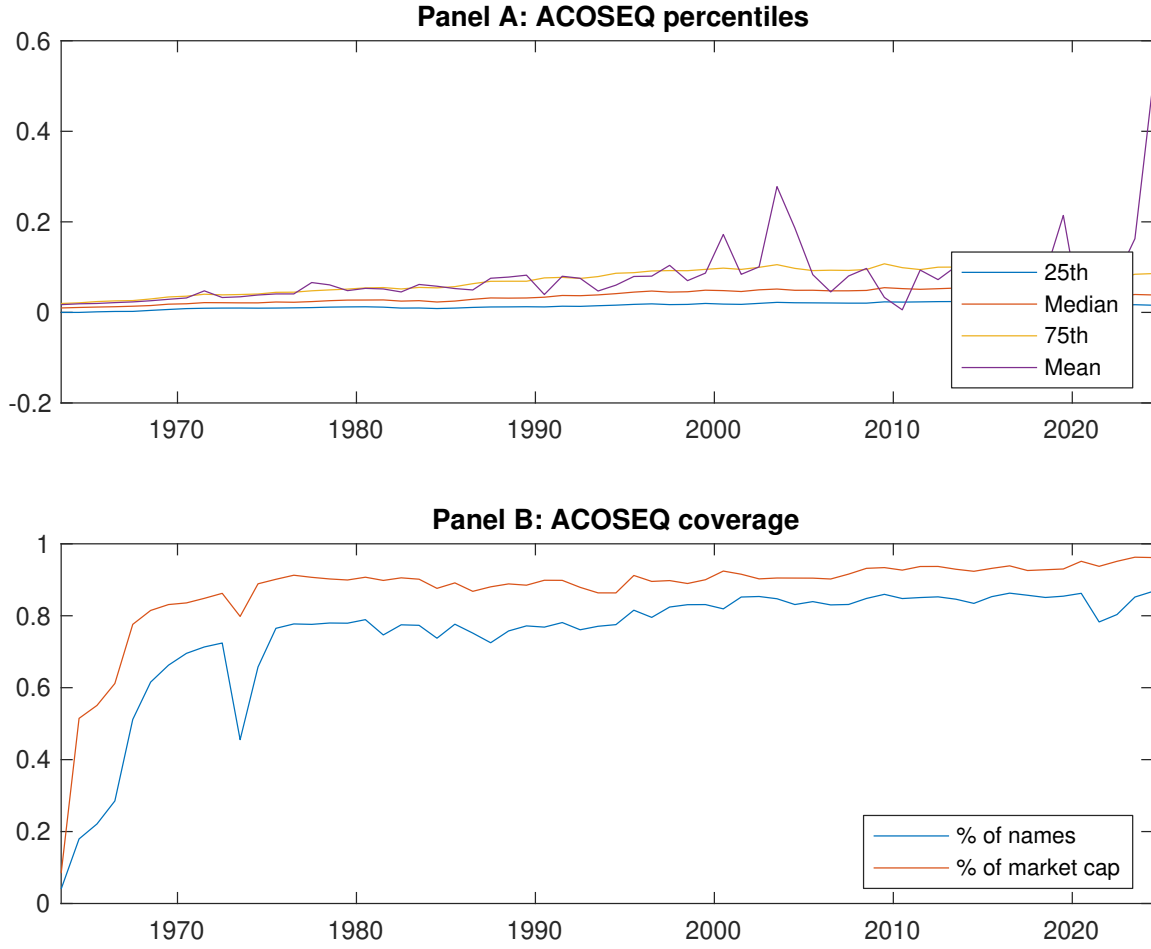


Figure 1: Times series of LLI percentiles and coverage. This figure plots descriptive statistics for LLI. Panel A shows cross-sectional percentiles of LLI over the sample. Panel B plots the monthly coverage of LLI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on LLI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.80]	0.62 [3.58]	0.54 [3.27]	0.66 [3.96]	0.76 [4.34]	0.25 [3.01]
α_{CAPM}	-0.11 [-1.89]	0.03 [0.57]	-0.03 [-0.60]	0.09 [1.96]	0.16 [3.37]	0.27 [3.13]
α_{FF3}	-0.19 [-4.10]	0.06 [1.17]	0.01 [0.26]	0.13 [3.05]	0.18 [3.74]	0.37 [4.83]
α_{FF4}	-0.16 [-3.37]	0.04 [0.71]	0.04 [0.90]	0.11 [2.57]	0.16 [3.41]	0.32 [4.19]
α_{FF5}	-0.15 [-3.19]	0.10 [1.90]	0.01 [0.30]	0.06 [1.38]	0.08 [1.87]	0.24 [3.19]
α_{FF6}	-0.13 [-2.66]	0.07 [1.43]	0.04 [0.84]	0.05 [1.13]	0.08 [1.78]	0.21 [2.79]
Panel B: Fama and French (2018) 6-factor model loadings for LLI-sorted portfolios						
β_{MKT}	1.08 [92.79]	0.99 [81.43]	0.94 [87.61]	0.98 [96.00]	1.02 [93.27]	-0.06 [-3.26]
β_{SMB}	-0.12 [-7.01]	-0.00 [-0.20]	0.03 [1.64]	-0.00 [-0.02]	0.11 [6.98]	0.23 [8.74]
β_{HML}	0.28 [12.51]	-0.03 [-1.29]	-0.13 [-6.14]	-0.15 [-7.67]	-0.11 [-5.27]	-0.39 [-11.26]
β_{RMW}	-0.09 [-4.03]	-0.07 [-2.89]	0.01 [0.31]	0.16 [7.98]	0.22 [10.19]	0.31 [8.77]
β_{CMA}	-0.04 [-1.18]	-0.09 [-2.74]	0.00 [0.13]	0.09 [2.95]	0.10 [3.06]	0.13 [2.61]
β_{UMD}	-0.03 [-2.96]	0.03 [2.68]	-0.03 [-3.24]	0.01 [1.41]	0.00 [0.38]	0.04 [2.13]
Panel C: Average number of firms (n) and market capitalization (me)						
n	711	795	690	608	601	
me (\$10 ⁶)	1616	1882	1910	2219	2051	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LLI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.01]	0.27 [3.13]	0.37 [4.83]	0.32 [4.19]	0.24 [3.19]	0.21 [2.79]
Quintile	NYSE	EW	0.17 [2.59]	0.12 [1.97]	0.10 [1.66]	0.11 [1.72]	0.00 [0.06]	0.02 [0.27]
Quintile	Name	VW	0.26 [3.03]	0.28 [3.18]	0.39 [4.88]	0.34 [4.19]	0.24 [3.16]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.44]	0.23 [2.78]	0.32 [4.21]	0.30 [3.88]	0.18 [2.44]	0.17 [2.35]
Decile	NYSE	VW	0.18 [1.76]	0.20 [1.92]	0.30 [3.29]	0.32 [3.45]	0.22 [2.37]	0.25 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.77]	0.25 [2.92]	0.34 [4.42]	0.32 [4.09]	0.22 [2.97]	0.20 [2.76]
Quintile	NYSE	EW	0.01 [0.08]					
Quintile	Name	VW	0.24 [2.81]	0.26 [2.99]	0.36 [4.50]	0.33 [4.13]	0.22 [2.98]	0.21 [2.74]
Quintile	Cap	VW	0.18 [2.19]	0.20 [2.50]	0.28 [3.76]	0.27 [3.60]	0.16 [2.20]	0.16 [2.15]
Decile	NYSE	VW	0.15 [1.51]	0.18 [1.70]	0.27 [2.91]	0.28 [3.03]	0.18 [2.02]	0.20 [2.16]

Table 3: Conditional sort on size and LLI

This table presents results for conditional double sorts on size and LLI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLI and short stocks with low LLI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LLI Quintiles					LLI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.54 [2.20]	0.70 [2.67]	0.81 [3.22]	0.91 [3.44]	0.73 [2.73]	0.19 [2.13]	0.13 [1.48]	0.08 [0.94]	0.11 [1.22]	0.04 [0.40]	0.06 [0.69]
	(2)	0.65 [2.86]	0.75 [3.13]	0.82 [3.51]	0.73 [3.03]	0.89 [3.70]	0.24 [2.64]	0.19 [2.19]	0.17 [1.87]	0.21 [2.35]	0.08 [0.86]	0.12 [1.36]
	(3)	0.76 [3.75]	0.64 [3.04]	0.77 [3.54]	0.88 [3.92]	0.87 [3.88]	0.12 [1.26]	0.05 [0.57]	0.03 [0.30]	0.04 [0.38]	-0.06 [-0.64]	-0.05 [-0.48]
	(4)	0.60 [3.08]	0.57 [2.91]	0.75 [3.83]	0.86 [4.32]	0.78 [3.84]	0.18 [2.09]	0.15 [1.76]	0.21 [2.53]	0.20 [2.38]	0.11 [1.33]	0.11 [1.33]
	(5)	0.50 [2.71]	0.60 [3.64]	0.47 [2.89]	0.61 [3.78]	0.72 [4.27]	0.23 [2.35]	0.26 [2.71]	0.39 [4.48]	0.35 [3.90]	0.25 [2.93]	0.22 [2.58]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LLI Quintiles					LLI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	380	389	387	386	379	31	32	32	31	28	
	(2)	104	105	105	104	104	54	54	55	54	54	
	(3)	74	74	74	74	74	94	92	95	95	96	
	(4)	62	62	62	62	62	204	203	205	205	212	
(5)	56	56	56	56	56	1354	1543	1539	1700	1617		

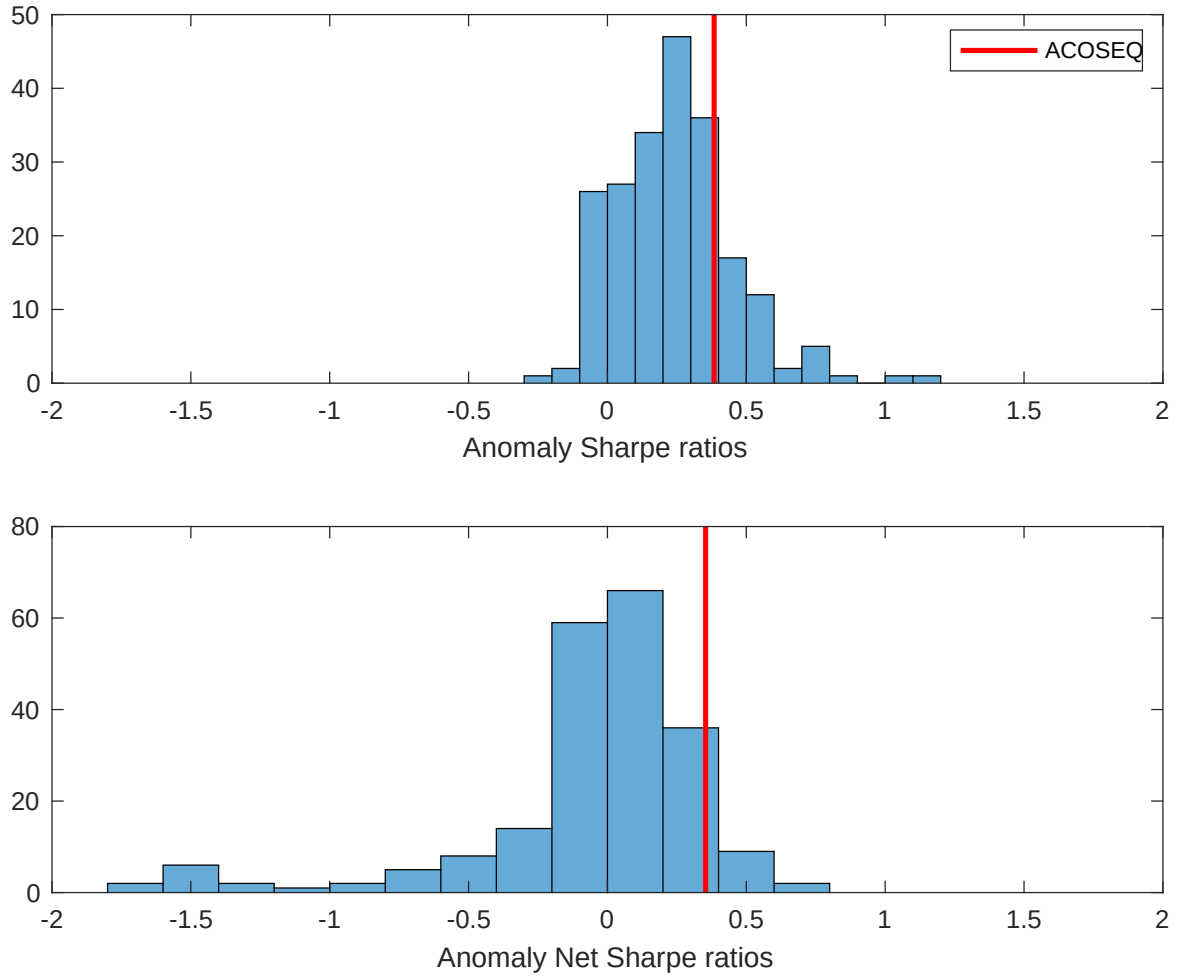


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

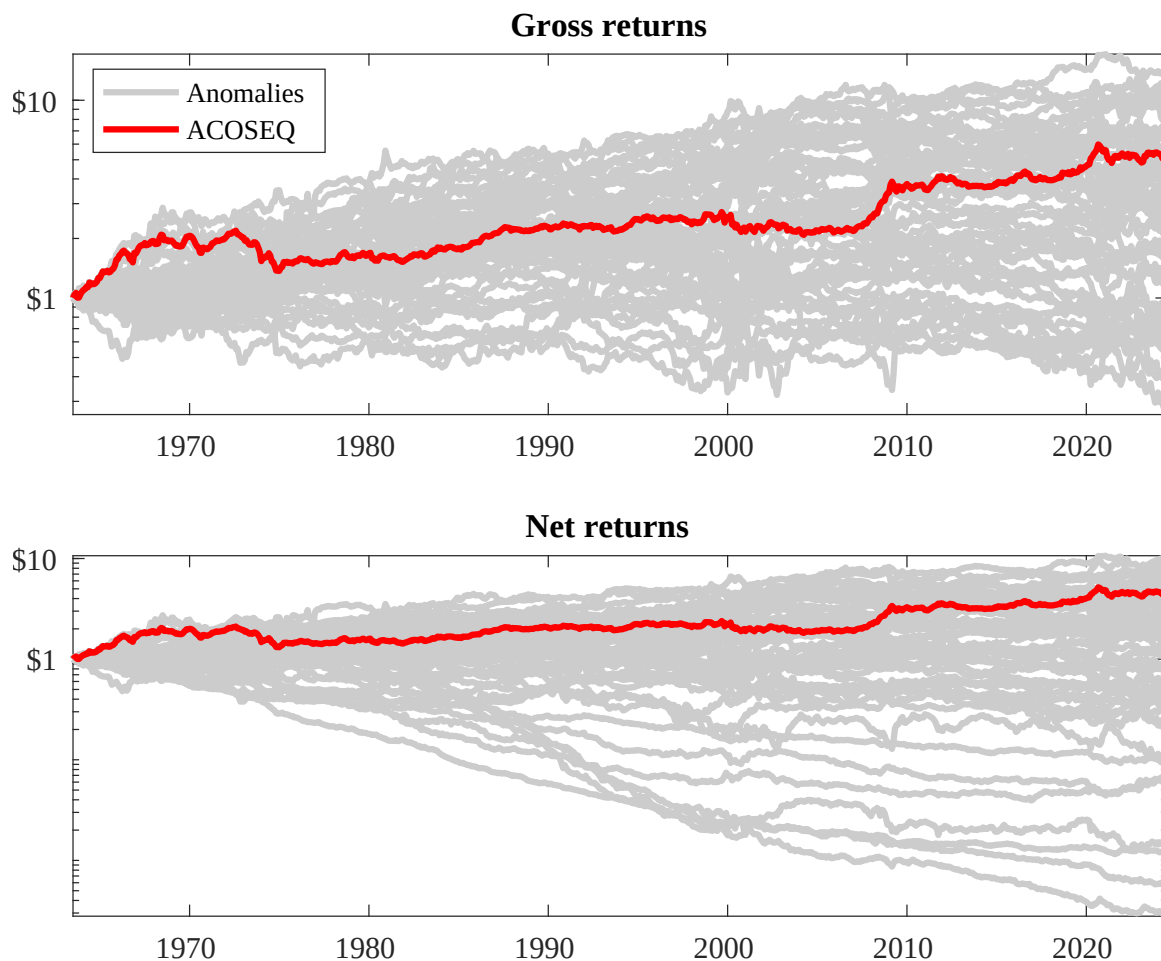
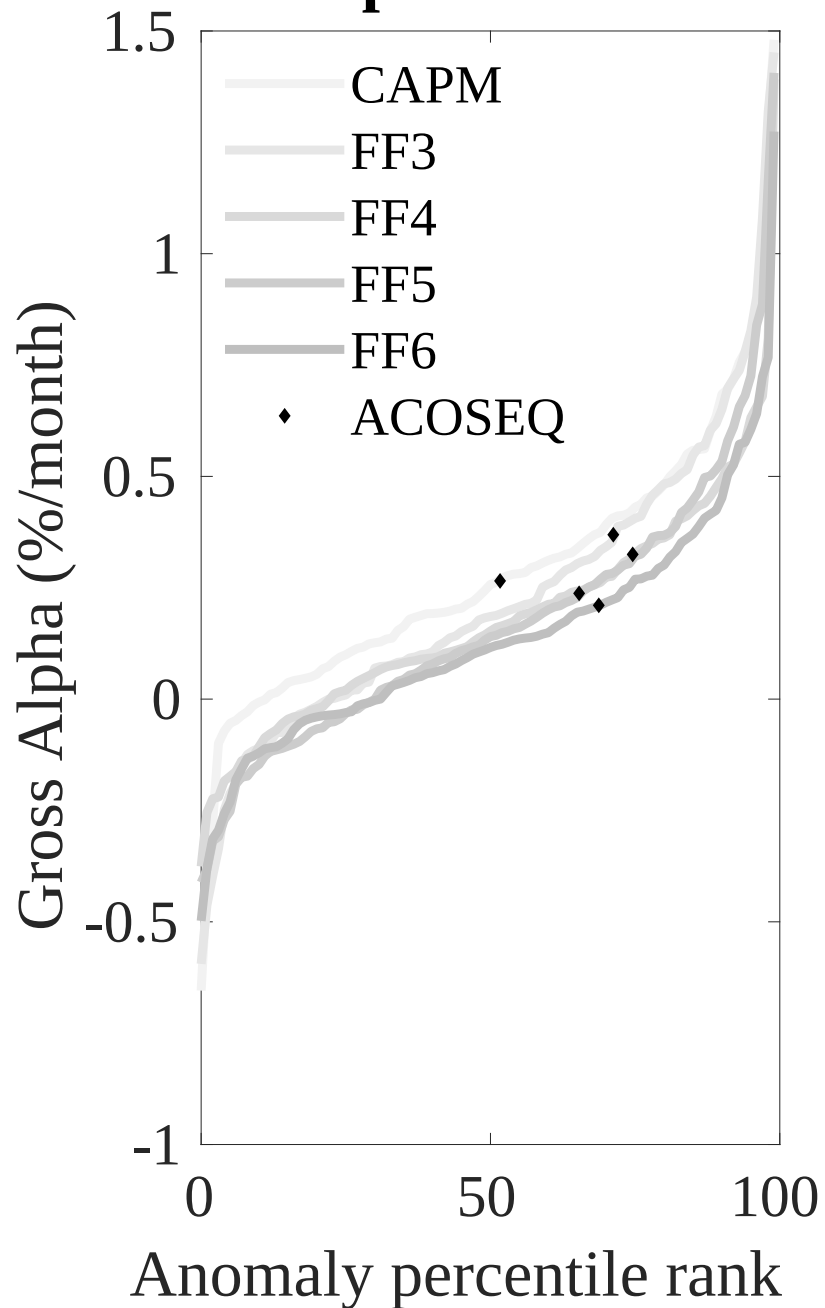


Figure 3: Dollar invested.

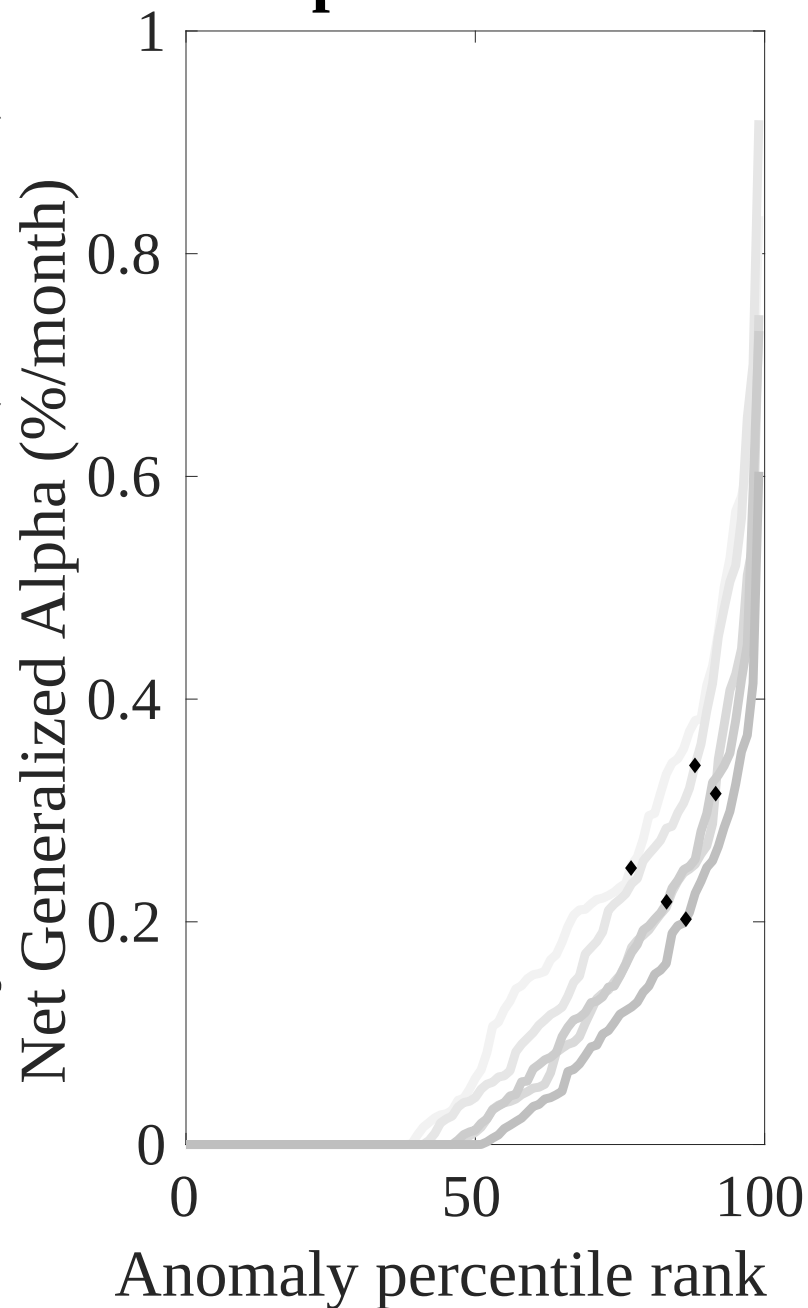
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

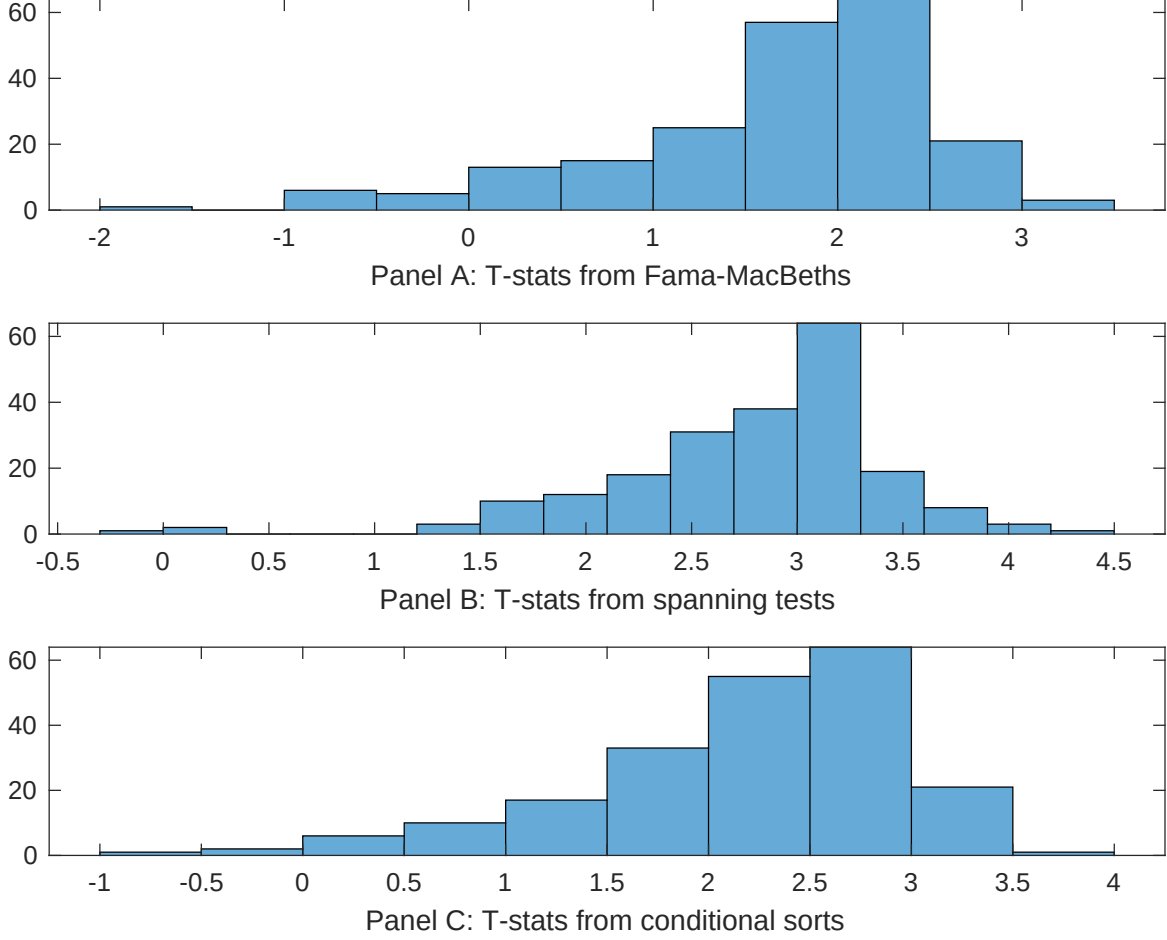


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LLI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [4.40]	0.92 [3.70]	0.14 [6.25]	0.12 [5.09]	0.11 [4.81]	0.20 [9.08]	0.17 [7.61]
LLI	0.57 [1.60]	0.67 [1.75]	0.12 [2.94]	0.90 [2.75]	0.82 [2.24]	0.11 [3.10]	0.69 [2.33]
Anomaly 1	0.13 [3.58]						0.66 [0.02]
Anomaly 2		0.68 [4.90]					0.64 [4.77]
Anomaly 3			0.37 [7.37]				0.26 [4.66]
Anomaly 4				0.13 [1.31]			-0.43 [-1.83]
Anomaly 5					0.60 [4.79]		0.36 [2.32]
Anomaly 6						0.51 [5.26]	0.43 [3.93]
# months	720	720	715	720	715	720	715
$\bar{R}^2(\%)$	0	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [2.38]	0.14 [1.85]	0.19 [2.48]	0.18 [2.40]	0.25 [3.38]	0.20 [2.70]	0.17 [2.37]
Anomaly 1	27.99 [9.78]						22.12 [6.95]
Anomaly 2		21.69 [7.21]					14.05 [4.07]
Anomaly 3			-19.68 [-4.53]				-10.70 [-2.30]
Anomaly 4				10.37 [3.70]			0.96 [0.35]
Anomaly 5					-17.13 [-4.65]		-20.41 [-5.65]
Anomaly 6						-10.30 [-2.65]	4.57 [1.10]
mkt	-5.57 [-3.27]	-6.19 [-3.54]	-5.14 [-2.86]	-4.52 [-2.47]	-4.18 [-2.30]	-5.72 [-3.17]	-3.41 [-1.99]
smb	10.73 [3.86]	21.90 [8.63]	30.73 [10.09]	22.67 [8.72]	24.69 [9.50]	26.26 [9.25]	16.64 [5.28]
hml	-23.67 [-6.53]	-29.14 [-8.04]	-18.08 [-3.23]	-31.41 [-7.94]	-28.66 [-7.16]	-26.90 [-4.82]	-2.85 [-0.48]
rmw	15.34 [4.17]	18.23 [4.76]	27.74 [7.81]	33.88 [9.39]	24.99 [6.76]	30.48 [8.66]	2.28 [0.53]
cma	7.40 [1.52]	16.22 [3.25]	14.48 [2.84]	12.81 [2.51]	9.63 [1.88]	10.74 [2.07]	9.04 [1.84]
umd	3.41 [2.02]	3.36 [1.93]	4.91 [2.75]	3.51 [1.96]	0.18 [0.09]	3.85 [2.15]	-0.87 [-0.47]
# months	732	732	728	732	728	732	728
$\bar{R}^2(\%)$	36	33	30	29	30	29	41

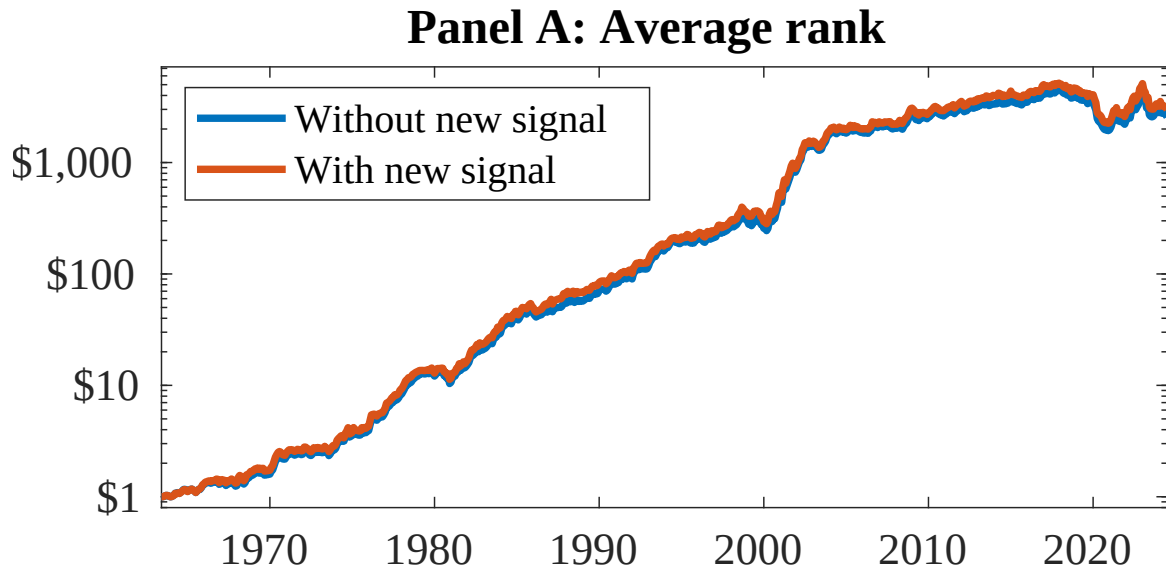


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as LLI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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